Solution report

Written Solution Report

Cover Section

Field	Details
Team Name	*Code-Blooded*
Institution Name	*Drs. Kiran & Pallavi Patel Global University*
Yi Chapter	*Vadodara*
City	*Vadodara*
Track	Accessibility
Problem Statement Title	Audit to Action
Solution Name	**Sarvasya — Access for All**
Team Members & Roles	*Hetanshi Sidhpura (Project Administrator) - Captain* *Aaryan Jaiswal (Tech Lead)* *Dhruvi Jamnapara (Research)*
Date of Chapter Level Round	*22nd August 2026*

3.1 Problem Understanding

India is home to over **2.68 crore persons with disabilities** (Census 2011), though according to WHO the estimate can read upto 7 to 11 crore people.India has laws like the RPwD Act 2016 & programs like Sugamya Bharat Abhiyan to help people with disabilities. However an audit by the NCPEDP found that less than 3% of government buildings are fully accessible to people with disabilities & essential places like courts & hospitals still remain of reach.
This failure stems from four critical systemic gaps:

- **Zero transparency and accountability:** There is no publicly accessible, real-time database that tracks which buildings are compliant & verify before visiting or for officials to be held accountable.
- **Absence of a structured complaint-to-resolution pipeline:** Even when barriers are reported, compliants are either left unactioned without tracking & assigned officers or dismissed without any valid justification.
- **Fragmented community infrastructure:** Volunteers, NGOs, caregivers, & especially-abled persons operate in private with no unified platform to coordinate physical help or emergency aid.
- **No incentives:** Building owners and municipal bodies face negligible consequences for non-compliance because the audit results are not public, & citizens lack the tools to demand better.

The problem is the **lack of accountibility and information(transparency)**.

3.2 Proposed Solution

Solution Name: Sarvasya — Access for All

One-line Description: Sarvasya is a public platform that ensures Indian buildings are accessible for persons with disabilities by transparently tracking building compliance audits, displaying live accessibility ratings, and providing built-in indoor navigation and emergency safety tools..

Solution Type: ☑ Product · ☑ Technology · ☑ Platform · ☑ Service · ☐ Hardware · ☐ Hybrid

How It Works — Core Features

Sarvasya is an application (designed to work offline via localStorage caching) that addresses accessibility at three interconnected levels: **Transparency, Accountability, and Community Action**

Layer 1: Transparent Building Directory & Ratings

Every public building in the platform's registry displays a **live accessibility rating** calculated from official audit scores, community-submitted field reports, and unresolved complaint counts. Buildings are categorised (Hospitals, Government Offices, Libraries) with hospitals prioritised as critical institutions. Citizens can search, filter by compliance status (Compliant / Needs Attention / Action Required), and view detailed records including:

- Accessible features (ramps, tactile paths, Braille lifts, grab rails)
- Compliance gap reports with specific references to NBC 2016 and RPwD Act clauses
- Independently verified field audit summaries with auditor names
- Indoor wayfinding maps with checkpoint statuses

This public visibility creates **reputational pressure** — the root cause of non-compliance is that nobody was watching. Now everyone can see.

Layer 2: Transparent Complaint Pipeline with Penalties

Citizens can file accessibility complaints against specific buildings. Each complaint enters a **shipping-tracker-style pipeline** visible to the complainant:

Submitted → Assigned to Officer (name visible) → In Progress → Resolved / Dismissed

- The **assigned officer's name** is publicly displayed, encouraging personal accountability.
- If a complaint is **dismissed**, the officer must provide a **mandatory written justification** visible to the user, maintaining transparency.
- Users who file **3 or more complaints flagged as fake** are **locked out** from further submissions — deterring misuse while protecting system integrity.

This directly addresses the root cause of complaints disappearing into opaque bureaucracy.

Layer 3: Safety & Community Assistance

- **Emergency Button**: A persistent, always-visible button that silently broadcasts the user's indoor location to emergency contacts, building security, and nearby volunteers — without causing public panic.
- **Caregiver Remote Tracking**: A toggle that lets a family member or caregiver monitor safe passage through complex buildings, with automatic alerts if the user gets stuck.
- **Find a Buddy**: Broadcasts a help request to verified nearby volunteers and security personnel, showing their distance and role.
- **Voice Command Mode**: Hands-free operation including a simulated sign-reader that reads boards and signage aloud.
- **Safe Spots**: Users can save and instantly retrieve evacuation-safe zones (fire exits, wheelchair refuges) as shortcuts.

Layer 4: Volunteering & Community Action

- Browse NGOs, orphanages, old-age homes (Vrudhashrams), and special schools.
- Book occasion slots (birthdays, anniversaries) to spend with residents.
- Donate directly through the platform.
- Sign up for volunteer tasks (audit assistance, reading, meal service).

Layer 5: Accessibility Controls & Offline Support

- **Variable text size** (A-, A, A+), **high contrast**, **colour inversion**, and **colorblind-safe themes** (Deuteranopia/Tritanopia) built into every screen.
- **Screen reader compatibility** with ARIA labels, semantic HTML, skip-to-content links, and keyboard navigation.
- **Offline-first architecture**: All data persists in localStorage, enabling use in areas with poor connectivity.

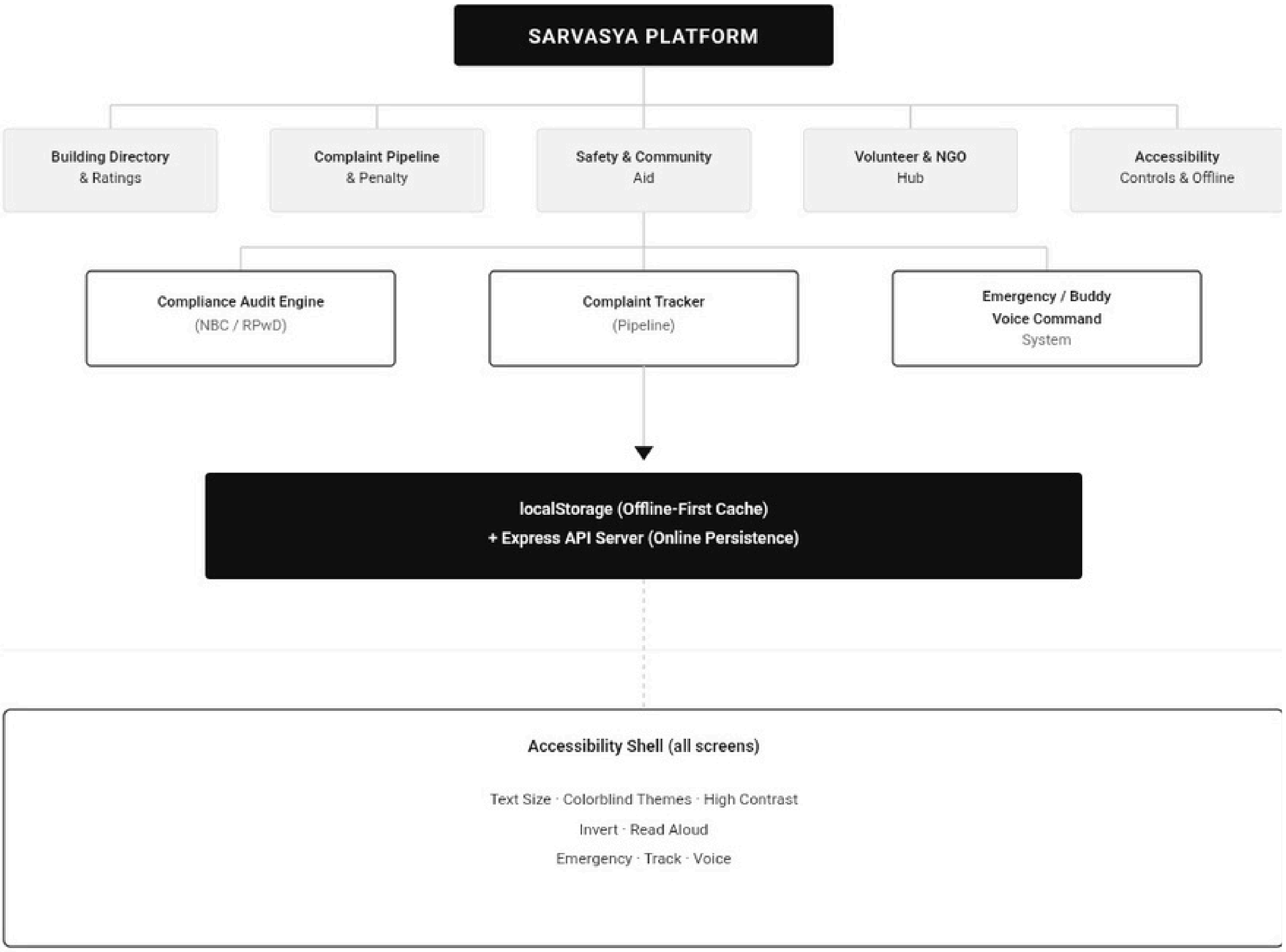
How This Addresses the **Root Causes** (from 3.1)

Root Cause	Sarvasya's Response
No transparency	Public building directory with live ratings and audit trails
No complaint accountability	Shipping-tracker pipeline with named officers and mandatory dismiss reasons
Fragmented community aid	Emergency button, buddy system, caregiver tracking, volunteering hub
No incentive for compliance	Public ratings create reputational pressure; complaint data feeds ratings

Target Users / Beneficiaries

- **Primary**: Persons with disabilities (mobility, visual, cognitive) navigating public buildings
- **Secondary**: Caregivers, family members, elderly citizens, pregnant women
- **Tertiary**: Municipal officials, architects, auditors, NGOs, volunteers

System architecture:



3.4 Feasibility & Scalability [20 marks]

Implementation Plan — First 90 Days

Days	Milestone
1–15	Deploy pilot in one municipal ward (e.g., Pune Ward 15 — Shivajinagar). Onboard 10 public buildings, 3 local NGOs, and 1 municipal accessibility officer.
16–30	Conduct on-ground accessibility audits for pilot buildings with partner NGOs. Populate compliance data and wayfinding maps.
31–50	Launch public-facing platform with pilot data. Distribute QR-coded posters at building entrances linking to their Sarvasya record. Onboard 50 citizen testers including wheelchair users and visually impaired persons.
51–75	Collect complaint pipeline data. Train 2 municipal officers on the complaint resolution workflow. Iterate based on user feedback — especially assistive features.
76–90	Publish pilot impact report: buildings audited, complaints resolved, average resolution time, user satisfaction. Prepare expansion proposal for city-level rollout.

Rough Cost Estimate:-

Item	Estimated Cost (INR)
Cloud hosting (1 year, low-traffic tier)	₹36,000
On-ground audit kits (measuring tape, inclinometer, clipboard) × 5	₹5,000
QR posters and printed awareness material	₹8,000
Travel and field expenses (pilot ward, 90 days)	₹15,000
Volunteer stipends (3 field auditors × ₹3,000/month × 3 months)	₹27,000
Miscellaneous (SIM data packs, printing, contingency)	₹9,000
Total Pilot Cost	**₹1,00,000 (approx.)**

This is deliberately bootstrapped. The platform itself is built using open-source tools (React, Vite, Express) with zero licensing costs.

Required Infrastructure, Resources, and Partners

- **Infrastructure**: A laptop, internet connection, and a ₹3,000/year cloud server. The platform already runs offline.
- **Partners**: Local disability rights NGOs (e.g., NCPEDP local chapters, Samarthyam, AccessAbility), municipal ward offices, Yi chapter volunteers.
- **Resources**: Accessibility auditing training (free manuals available from CPWD and MoHUA), smartphone with camera for field audits.

Key Stakeholders and Roles

Stakeholder	Role
Municipal Ward Officer	Assigns complaints, validates audit results
Disability Rights NGOs	Conduct field audits, recruit testers
Citizen Volunteers	File reports, use buddy system, verify buildings
Architects / Builders	Use compliance checker before construction
Yi Chapter	Coordinate pilot, community outreach

Scalability Roadmap

Pilot (1 Ward, 10 buildings)

↓ 90 days

City (1 City, 100+ buildings)

↓ 6 months — partner with Smart Cities Mission

State (Multiple cities, 1,000+ buildings)

↓ 12 months — integrate with state urban development portals

National (All states, 10,000+ buildings)

↓ 24 months — API integration with MoHUA's national building database

3.5 Innovation & Differentiation

What makes Sarvasya stand out.

Most existing accessibility efforts in India are **report-based,** an NGO audits a building, files a report, and the report gathers dust. Sarvasya inverts this model by making accessibility data **live, online, public, and actionable**.

Comparison With Existing Solutions:

Existing Effort	What It Misses	What Sarvasya Adds
Sugamya Bharat Abhiyan (govt campaign)	Top-down, no citizen feedback loop, no public data	Bottom-up citizen reporting + public transparency
Wheelmap.org (global)	Only marks wheelchair accessibility; no compliance data, no complaints	Full compliance scoring, complaint pipeline, Indian building codes
AccessNow (app)	Crowdsourced reviews only; no accountability pipeline	Named officer assignment, penalty system, mandatory dismiss justification
Manual CPWD audits	Paper-based, no public disclosure, no follow-up.	Digital audit trail, real-time ratings, wayfinding maps

Unique Approach

The **complaint-to-rating feedback loop** is our core innovation: every unresolved complaint dynamically reduces a building's public rating. This creates a self-reinforcing accountability engine, buildings that ignore complaints see their ratings drop publicly, creating pressure that no internal report ever could.
The Ai in our application performs the first verification of the document and information uploded by the builder, and once accepted by it, it's forwarded to an inspector to perform an audit review.
The complaints sent by the users are processed in a transperant way compared to the existing applications where the users can see how and where their complaints are being processed.

3.6 Impact Potential

Who Benefits, and How Many

Direct beneficiaries: India's **2.68 crore** registered persons with disabilities, plus an estimated **4 - 8 crore** unregistered (WHO methodology). In a single pilot city like Pune (population ~37 lakh), approximately **1.5 - 3 lakh persons with disabilities** stand to benefit.
Indirect beneficiaries: Elderly citizens (~**13.8 crore**, Census 2011 projection for 2026), pregnant women, persons with temporary injuries, and parents with strollers, all of whom benefit from barrier-free infrastructure.
Civic impact: Every complaint resolved means one fewer barrier for thousands of future visitors to that building.

Measurable KPIs

KPI	Target (Year 1 Pilot)
Buildings audited and publicly rated	100+
Accessibility complaints filed	500+
Average complaint resolution time	< 30 days
Complaints resolved vs. dismissed ratio	> 70% resolved
Volunteer buddy assists completed	200+
NGO occasion slots booked	50+
Emergency broadcasts triggered	Tracked (any non-zero = lives potentially saved)
User satisfaction (post-visit survey)	> 80% positive

Expected Timeframe for Real-World Impact

- **30 days**: First buildings publicly rated; first complaints in the pipeline.
- **90 days**: Measurable complaint resolution data; municipal officers actively using the system.
- **6 months**: Demonstrable building compliance improvements in the pilot ward driven by public rating pressure.
- **12 months**: City-level data sufficient to present to state government for policy integration.

Sarvasya does not wait for policy to change the world but rather it creates the data and pressure that makes policy change inevitable.